

COMPDYN 2027

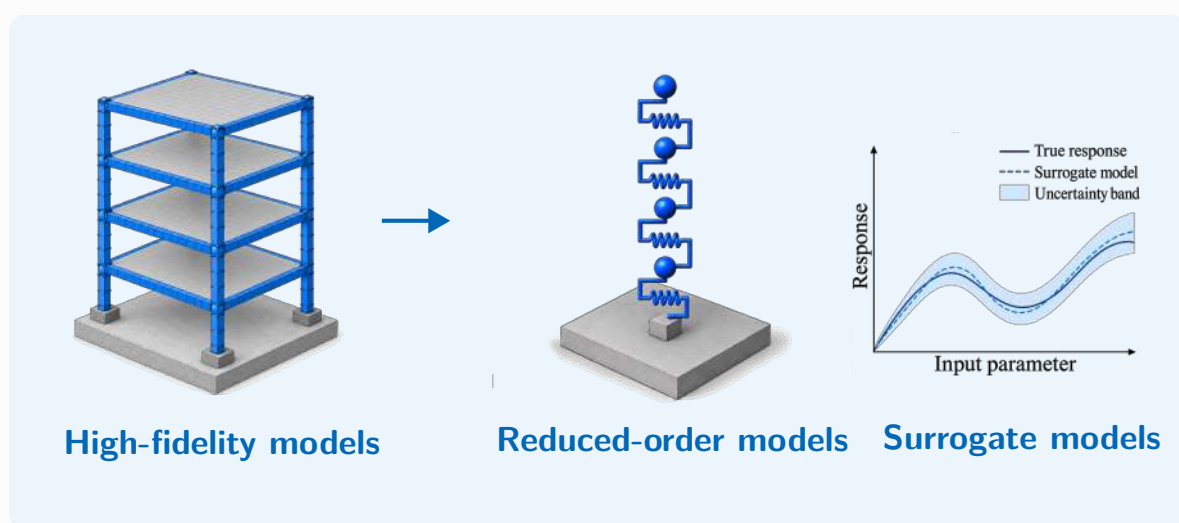
11th International Conference on Computational
Methods in Structural Dynamics and Earthquake Engineering

Mini-Symposium

Reduced-Order and Surrogate Models

Advances in Computationally Efficient Structural Dynamics

Probabilistic structural assessment often requires extensive **nonlinear dynamic analyses** to account for uncertainties in natural hazards, structural properties, exposure, and capacity models. **Reduced-order** and **surrogate modelling strategies** offer computationally efficient alternatives while preserving response features needed for structural performance and risk analysis. **This mini-symposium** provides a forum for **recent advances** in model reduction, surrogate modelling, calibration, validation, and uncertainty quantification applications in structural dynamics



Topics include:

- Formulation and calibration of **reduced-order** and **surrogate models**
- Methods for accounting, or reducing for **model bias** and **uncertainties**
- **Validation** against high-fidelity numerical models and/or experimental data
- Computationally demanding applications, including **fragility analysis** and **regional-scale assessment**
- Case studies on structural response assessment under **seismic**, **tsunami**, **debris-flow**, and other **hazard scenarios**.

Organizers:

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Important dates

Abstract submission: **30 October 2026**
Abstract acceptance: **30 November 2026**
Full paper (optional) sub: **28 February 2027**
Full paper acceptance: **31 March 2027**

More info

2027.compdyn.org

